

Virtual Lab

Concepts & Visual Experiments — Class 1 to 12

Based on NCERT / CBSE Curriculum • All Subjects • NEP 2020 Aligned

Each experiment is designed to be visualised or demonstrated on an interactive digital board



How to Use This List

- Left column — the NCERT concept (topic from the textbook)
- Right column — a ready-to-use digital board experiment: each one is interactive, visual, and requires no physical lab equipment
- Tools that support these: GeoGebra, PhET Simulations (phet.colorado.edu), OLABs, Desmos, DIKSHA, or a custom school app

STAGES AT A GLANCE

Classes	Stage	Subjects Covered
1 – 2	Foundation	Mathematics, EVS, English
3 – 5	Preparatory	Mathematics, Science, Social Science, English
6 – 8	Middle	Mathematics, Science, Social Science, English
9 – 10	Secondary	Mathematics, Physics, Chemistry, Biology, Social Science
11 – 12	Senior Secondary	Maths, Physics, Chemistry, Biology, Accountancy, Economics, History, Geography, Political Science, CS

CLASS 1 – 2 | Foundation Stage

Mathematics



Concept



Digital Board Experiment / Visualisation

Counting & Numbers (1–100)	Animated bead abacus: click beads to build numbers; show tens and ones columns live on screen
Shapes & Space	Drag 2D shapes (circle, square, triangle, rectangle) into matching silhouettes with pop animation on correct placement
Addition & Subtraction	Apple-tree counter: tap to add apples / remove apples; running total updates in large numerals
Patterns (colour, shape)	Pattern strip builder: identify the missing item in an AB, ABB or ABC repeating pattern; highlight rule on completion
Measurement (big/small, long/short)	Side-by-side comparison tool: drag two objects next to each other, an animated arrow labels the longer / shorter one
Money Basics	Virtual coin tray: drag ₹1, ₹2, ₹5 coins to match a target amount; real-time running total shown

Environmental Studies (EVS)

 Concept	 Digital Board Experiment / Visualisation
My Family & Relationships	Interactive family-tree canvas: place family-member photos/icons and draw lines to show relations
Animals & Their Habitats	Drag-and-drop habitat matcher: animals (lion, fish, eagle, frog) matched to forest / water / sky / pond with animated movement
Plants Around Us	Virtual plant growth: tap Water / Sun / Soil buttons to grow seed → sprout → sapling → flower step by step
Food We Eat	Balanced-plate builder: sort food cards (grains, vegetables, fruits, dairy, protein) onto a plate; plate 'smiles' when balanced
Water & Its Uses	Water-cycle loop animation: tap 'Sun' to evaporate, clouds form and rain falls, rivers flow — narrated loop
Our Senses	Tap sense-organ hotspots on a body diagram; short clip of sound/smell/taste/sight/touch activates with vibration icon

English

 Concept	 Digital Board Experiment / Visualisation
Alphabet Recognition & Phonics	Letter-tap board: tap each letter, hear its phoneme, matching picture bounces in (A-Apple, B-Ball)
CVC Word Building	Drag letter tiles into 3-slot frame; when a real word forms, a picture appears and audio plays
Rhyming Words	Rhyme-pair flip game: cards fly together with a 'ping' sound when a rhyming pair is matched

CLASS 3 – 5 | Preparatory Stage

Mathematics

 Concept	 Digital Board Experiment / Visualisation
Large Numbers & Place Value	Digital abacus (up to 99,999): slide beads in ones/tens/hundreds/thousands/ten-thousands columns; number updates instantly
Fractions	Shape-slicer: divide a circle/rectangle into equal parts, shade the numerator fraction; fraction notation appears below
Multiplication & Division	Area-model grid: draw a rectangle $W \times H$, fill each cell, count total squares to see multiplication as area
Geometry – 2D & 3D Shapes	3D shape viewer: rotate cube, cone, cylinder, sphere; press 'Unfold' to see the net flat on screen
Perimeter & Area	Dot-paper canvas: draw shapes by connecting dots; perimeter and area calculated and displayed in real time
Data Handling – Pictograph & Bar Graph	Class-survey tool: enter votes for favourite fruit/colour; bar graph animates upward bar-by-bar; swap to pictograph view
Money & Mental Math	Virtual market: pick items with price tags, count coins/notes to pay, animated change returned

Science

 Concept	 Digital Board Experiment / Visualisation
Food & Nutrition	Balanced-plate designer: drag food groups into plate zones; nutritional bar (protein, carbs, vitamins) fills or warns if unbalanced
Living vs Non-living Things	Sort cards onto 'Living' / 'Non-living' columns; correct drops glow green, incorrect shake red — score shown
Adaptation of Animals & Plants	Environment selector (desert / Arctic / rainforest): watch animal/plant features highlight that help survival
Force & Motion	Ball-push simulator: set 'Push Strength' slider; ball rolls farther on smooth vs rough surface — friction visualised
States of Matter	Heat slider: move temperature dial, watch particle animation change from solid → liquid → gas with phase labels
Simple Machines (Lever, Pulley)	Interactive lever: adjust fulcrum position and effort arm length; see load side rise/fall with effort calculation shown

Social Science

 Concept	 Digital Board Experiment / Visualisation
Maps – Directions & Symbols	Zoomable map canvas: tap compass points to orient; match map symbols (river, mountain, road) to legend items

Our Earth – Landforms & Water Bodies	3D globe: click landform (mountain, plateau, plain, valley); animated cross-section and label appears
Natural & Man-made Resources	Resource-sorting board: drag cards into 'Natural' / 'Man-made' bins; timer adds challenge mode
Indian History – Ancient Civilisations	Timeline strip: tap periods (Indus Valley, Vedic, Maurya); map highlights region and key facts pop up

English

 Concept	 Digital Board Experiment / Visualisation
Parts of Speech	Colour-coded sentence builder: drag word cards; nouns turn blue, verbs red, adjectives green on correct placement
Reading Comprehension	Passage displayed with tap-to-highlight tool; students mark key ideas; summary auto-built from highlights
Tenses	Timeline ruler: drag sentence to Past / Present / Future zone; verb form updates automatically on correct placement

CLASS 6 – 8 | Middle Stage

Mathematics

 Concept	 Digital Board Experiment / Visualisation
Integers & Number Line	Animated number line: drag a pointer to add / subtract integers; jumps forward (positive) or backward (negative) shown with arrows
Fractions & Decimals	Fraction-to-decimal converter with visual: fraction bar splits, decimal value streams right; LCM / HCF finder with factor tree animation
Ratio & Proportion	Scaling workshop: input a recipe or map ratio; drag slider to scale up/down; bar comparison updates in real time
Algebraic Expressions	Algebra tiles: coloured square/rectangle tiles represent x^2 , x and constants; drag to add/subtract, like terms merge with animation
Linear Equations (one variable)	Balance-scale solver: move weights (terms) across the scale; scale tips until equation is balanced at the solution
Geometry – Lines, Angles, Triangles	Dynamic geometry tool: drag vertices of a triangle; all three angles update live and always sum to 180°
Perimeter, Area & Volume	3D box builder: input $L \times W \times H$, surface net unfolds; surface area and volume update as dimensions are dragged
Data Handling – Mean, Median, Mode	Number-set editor: add/remove values; mean, median, mode recalculate and are highlighted on a dot plot instantly
Percentage	Percentage visual: a 10×10 grid shades the given %; slider changes shading and shows fraction and decimal equivalents

Science

 Concept	 Digital Board Experiment / Visualisation
Cell – Basic Unit of Life (Cl. 8)	Zoomable cell diagram: tap organelles (nucleus, mitochondria, chloroplast); 3D pop-up with function label appears
Light – Reflection & Refraction (Cl. 8)	Ray-tracing canvas: move a mirror or glass slab; incident / reflected / refracted rays animate and angle values update
Electricity – Basic Circuit (Cl. 6)	Virtual circuit builder: drag bulb, battery, switch, wire; circuit glows when complete; short-circuit sparks as warning
Motion & Force (Cl. 7-8)	Distance-time graph plotter: move an animated car; graph draws itself; identify uniform vs non-uniform motion from slope
Sound – Propagation (Cl. 8)	Particle vibration animation: tap tuning fork, watch longitudinal wave (compression/rarefaction) travel through particles
Microorganisms (Cl. 8)	Microscope simulator: focus dial zooms into a slide; identify bacteria / fungi / protozoa with label-reveal on tap
Crop Production & Management (Cl. 8)	Farming cycle board: drag tools / processes (ploughing, sowing, irrigation, harvesting) onto calendar timeline

Conservation of Plants & Animals (Cl. 8)	Food-web builder: connect producer → herbivore → carnivore chains; remove one species, watch chain collapse visually
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Social Science

 Concept	 Digital Board Experiment / Visualisation
The Earth in the Solar System (Cl. 6)	Solar system orrery: planets orbit the sun at correct relative speeds; click each for size comparison and distance data
Major Landforms (Cl. 6)	Cross-section builder: layer mountains, plains, plateaus, valleys; label each with elevation band colour-coding
Water Bodies & Oceans (Cl. 6)	Interactive world map: tap ocean / sea names; animated current arrows appear; ocean depth colour gradient shown
Mughal Empire (Cl. 7)	Animated map: Mughal territory expands reign by reign with year counter; tap ruler name for portrait and key event
The Indian Constitution (Cl. 8)	Layered diagram of Preamble → Fundamental Rights → DPSP → Duties; tap layer to zoom into key articles
Resources & Development (Cl. 8)	Resource map of India: colour-coded by type (mineral, forest, water, soil); click region to see resource profile

English

 Concept	 Digital Board Experiment / Visualisation
Active & Passive Voice	Sentence flip card: show active sentence; tap 'Flip' to watch words rearrange into passive form with colour highlighting
Direct & Indirect Speech	Speech-bubble transformer: type direct speech into a bubble; tap 'Convert' — bubble reshapes into reported speech automatically
Letter / Essay Writing Structure	Document template with drag-drop section labels (salutation, body, conclusion); locked into correct order on completion

CLASS 9 – 10 | Secondary Stage

Mathematics

 Concept	 Digital Board Experiment / Visualisation
Number Systems (Real Numbers)	Number line magnifier: zoom into irrational segment between $\sqrt{2}$ and $\sqrt{3}$; show non-terminating decimal expansion live
Polynomials	Graph plotter: enter polynomial equation; curve draws live; drag to find roots where curve crosses x-axis — zero highlighted
Linear Equations in Two Variables	Coordinate canvas: input two equations; both lines draw simultaneously; intersection point (solution) glows and coordinates appear
Triangles – Congruence & Similarity	Triangle lab: adjust SSS / SAS / ASA conditions on two triangles; congruence rule fires a 'match' animation when satisfied
Circles – Theorems	Dynamic circle board: drag chord or tangent; angle values and theorem statements update in real time (angle in semicircle = 90°)
Coordinate Geometry	Drag-point canvas: place points, watch distance formula and midpoint calculate; quadrant labels update as points move
Statistics – Mean, Median, Mode of Grouped Data	Histogram builder: input frequency table; histogram draws; mean / median / mode marked with coloured lines on the graph
Probability	Coin / dice simulator: flip / roll 100 times; frequency graph builds; experimental probability vs theoretical shown side by side
Surface Area & Volume	3D solid explorer: enter radius / height; rotate the shape, colour-coded faces unfold to show surface; volume formula overlay
Trigonometry	Right-angle triangle scaler: drag hypotenuse angle slider; sin, cos, tan values update; unit circle overlay available

Physics

 Concept	 Digital Board Experiment / Visualisation
Motion – Distance-Time, Speed	Animated car on track: set speed, play; distance-time and speed-time graphs draw themselves; pause to read values
Laws of Motion (Newton's 3 Laws)	Collision simulator: set mass and velocity of two carts; watch elastic / inelastic collision; momentum calculated before/after
Gravitation	Planet gravity lab: select different planets; drop an object; observe different g values through different fall speeds and impact time
Work, Energy & Power	Inclined plane: push block up slope; energy bar chart splits into kinetic / potential / friction loss in real time
Light – Reflection & Mirrors	Mirror ray diagram: drag object near concave/convex mirror; image position, nature (real/virtual) and size update dynamically
Refraction – Lenses	Lens lab: drag object in front of convex/concave lens; real-time ray diagram draws and image type label appears

Electricity	Ohm's Law slider board: adjust V and R sliders; current I updates with graph; circuit diagram glows brighter with higher current
Magnetic Effects of Current	Field-line animator: place wire/solenoid on board; magnetic field lines appear and rotate when current direction is reversed

Chemistry

 Concept	 Digital Board Experiment / Visualisation
Atoms & Molecules	Element builder: drag protons, neutrons, electrons to build atoms; electron shell fills per Bohr model; element name reveals
Structure of Atom (Cl. 9)	Bohr model animator: select element; electrons orbit in shells; tap shell to see max electron count rule ($2n^2$)
Chemical Reactions & Equations	Equation balancer: drag atom tiles to balance left/right sides; scales tip until balanced; state symbols added on tap
Acids, Bases & Salts	pH scale strip: drag indicator into test tubes of different substances; colour changes across spectrum from red (acid) to blue (base)
Metals & Non-metals	Reactivity series tower: drag metals to rank order; drop water / acid on each — bubble intensity reflects reactivity
Carbon & its Compounds	Carbon chain builder: link carbon atoms; hydrogen atoms auto-fill; IUPAC name generates from structure built
Periodic Classification	Interactive periodic table: hover element — electron config, period, group, metallic nature highlight; filter by family

Biology

 Concept	 Digital Board Experiment / Visualisation
Cell Division – Mitosis & Meiosis	Stage-by-stage animation: tap NEXT through prophase → metaphase → anaphase → telophase with chromosome count shown
Tissues (Cl. 9)	Tissue type explorer: select plant/animal tissue; microscopic slide view appears; zoom in on cell arrangement with labels
Life Processes – Nutrition, Respiration	Body-system board: tap system (digestive/respiratory); animated pathway of food/air through organs with absorption zones lit
Reproduction (Cl. 10)	Flower cross-section: tap stamen/pistil to reveal gametes; watch pollination → fertilisation → seed formation step-by-step
Heredity & Evolution	Punnett square builder: input parent genotypes; offspring ratios fill into grid cells; phenotype percentage shown below
Our Environment – Food Chain & Web	Food-web canvas: connect species with arrows; remove a species — affected population sizes fluctuate visually (cascade effect)
Management of Natural Resources	Resource sustainability dial: set consumption rate vs replenishment; meter shows green/yellow/red for sustainability

Social Science

 Concept	 Digital Board Experiment / Visualisation
French Revolution	Animated timeline: key events pop up in sequence on a scrolling 1789–1799 strip with cause-effect arrows between events
Nazism & the Rise of Hitler	Propaganda-analysis board: show real poster images side by side with annotated labels identifying rhetorical techniques
India – Size & Location (Geography)	India outline map: tap states; latitude/longitude lines overlay; tropic of Cancer highlighted; distance scale shown
Drainage Systems of India	River system map: tap Himalayan rivers vs Peninsular rivers; watershed boundaries animate; tributaries trace upstream
Democratic Government – Parliament	Lawmaking flow diagram: Bill → Lok Sabha → Rajya Sabha → President → Act; each stage lights up on tap with explanation
Economic Development – Sectors	Sector pie chart: input GDP share data; chart redraws; tap each sector — examples and employment % appear as pop-up
Money & Banking	Bank transaction simulator: deposit, withdraw, take loan; account balance and interest tracker update in real time with ledger view

CLASS 11 – 12 | Senior Secondary Stage

Mathematics

 Concept	 Digital Board Experiment / Visualisation
Sets, Relations & Functions	Venn diagram builder: drag elements into sets A/B/C; Union, Intersection, Complement shade automatically as regions selected
Trigonometric Functions	Unit circle + graph sync: drag a point around the circle; $\sin \theta$ / $\cos \theta$ / $\tan \theta$ graphs draw simultaneously and values update live
Complex Numbers	Argand plane plotter: enter $a+bi$; point appears; modulus shown as radius, argument as angle arc — add two complex numbers with vector addition
Sequence & Series (AP / GP)	Term visualiser: enter first term and common difference/ratio; bars grow/shrink per term; sum displays as cumulative bar
Straight Lines & Conic Sections	Dynamic graph tool: input line/conic equation, shape draws; drag sliders to change a, b, c coefficients — shape transforms live
Limits & Derivatives (Cl. 11–12)	Tangent tracer: plot $f(x)$; drag a point along curve; tangent line rotates and slope value displays — show secant approaching tangent
Integration	Riemann sum visualiser: divide area under curve into n rectangles; increase n slider — rectangles shrink and area approaches exact integral
Vectors	3D vector canvas: input i, j, k components; vector arrow draws; add two vectors — resultant drawn with parallelogram law highlighted
3D Geometry	Axes rotator: place points in 3D, draw lines/planes; measure angle between planes; trace section through a solid
Probability – Conditional & Bayes	Tree diagram builder: enter branch probabilities; path multiplications auto-calculate; Bayes theorem box updates at end
Matrices & Determinants	Matrix operation pad: input two matrices; select Add/Multiply/Transpose/Inverse; step-by-step operation animates cell by cell
Linear Programming	Constraint plotter: input inequalities; feasible region shades; objective function line sweeps to corner point maximum/minimum

Physics

 Concept	 Digital Board Experiment / Visualisation
Kinematics – Projectile Motion	Launch angle simulator: set angle & speed, press Launch; parabolic path draws; max height, range, time values shown live

Rotational Motion & Torque	Turntable lab: set moment of inertia and angular velocity; change distribution of mass (slide weights outward); ω decreases in real time
Gravitation – Orbital Mechanics	Orbital simulator: place planet at distance r from star; orbit traces with Kepler's laws; $T^2 \propto r^3$ graph auto-plots
Fluid Mechanics – Bernoulli's Principle	Pipe flow visualiser: vary pipe diameter at different sections; velocity and pressure arrows update to show Bernoulli relation
Thermodynamics	PV diagram plotter: simulate isothermal / adiabatic / isochoric / isobaric processes; work done shown as shaded area under curve
Wave Optics – Interference & Diffraction	Young's Double Slit simulator: vary slit separation and wavelength; interference pattern (bright/dark fringes) redraws in real time
Electrostatics – Coulomb's Law	Charge interaction canvas: place positive/negative charges; electric field lines draw; force vectors update with Coulomb's formula
Current Electricity – Kirchhoff's Laws	Complex circuit board: build mesh; apply KCL/KVL; unknown currents solved and displayed as arrows in each branch
Electromagnetic Induction	Coil-magnet simulation: move bar magnet into/out of coil; galvanometer needle deflects; induced EMF graph plots in real time
Modern Physics – Photoelectric Effect	Frequency slider: vary light frequency hitting metal plate; current flows only above threshold; work function marked on graph
Semiconductor Devices – Diode / Transistor	IV characteristic plotter: sweep voltage on diode/transistor; characteristic curve draws; operating region highlighted
Nuclear Physics – Radioactive Decay	Decay simulator: N_0 atoms shown as dots; press Play; atoms 'pop' randomly; N-vs-t exponential curve builds; half-life marked

Chemistry

 Concept	 Digital Board Experiment / Visualisation
Mole Concept & Stoichiometry	Reaction mixer: input moles of reactants; stoichiometry calculator highlights limiting reagent; product moles shown with visual scale
States of Matter – Ideal Gas Laws	Gas container: adjust P, V, T sliders; particle animation speeds up/slows; $PV = nRT$ graph auto-updates
Atomic Structure – Quantum Numbers	Orbital shape viewer: select n, l quantum numbers; 3D s/p/d orbital shape renders; magnetic quantum number orientations shown
Chemical Bonding	Lewis structure builder: drag atoms, share electrons; bond type (single/double/triple/ionic) highlights on correct structure formed
Thermochemistry – Enthalpy	Energy diagram: draw Hess's law cycle; arrows pointing up (endothermic) or down (exothermic); net ΔH summed automatically
Chemical Kinetics	Rate-of-reaction graph: vary concentration/temperature/catalyst; reaction rate curve reshapes; activation energy E_a bar shown

Electrochemistry	Electrochemical cell builder: connect half-cell potentials from tables; EMF calculates; cell reaction writes itself; Nernst equation applied
Coordination Chemistry	Ligand-metal complex builder: drag ligands around metal centre; CFSE energy diagram draws for octahedral/tetrahedral splitting
Biomolecules	Molecule builder: assemble amino acids into a peptide chain; glucose into starch chain; 3D fold animation shows secondary structure

Biology

 Concept	 Digital Board Experiment / Visualisation
Plant Physiology – Photosynthesis	Chloroplast zoom: light-dependent vs Calvin cycle shown in two chambers; vary light intensity/CO ₂ ; output glucose/O ₂ rates update
Plant Physiology – Respiration	Mitochondria cross-section: glycolysis → Krebs cycle → ETC animated step by step; ATP counter increments at each stage
Animal Physiology – Digestion	Digestive tract journey: food item enters mouth; tap each organ, enzyme released, nutrient absorption highlighted at intestine wall
Animal Physiology – Circulation	Beating heart model: animate double circulation; oxygenated (red) vs deoxygenated (blue) blood flow; heart chambers labelled
Animal Physiology – Nervous System	Neuron signal propagator: tap sensory stimulus; action potential wave travels along axon; synaptic vesicles release at terminal
Genetics – Mendelian & Molecular	DNA replication animator: helicase unzips double helix; complementary bases snap into place; new strand highlighted
Biotechnology – Recombinant DNA	Gene cloning flowchart: cut DNA with restriction enzyme animation; insert into plasmid vector; transformation into bacteria shown
Ecology – Ecosystem Dynamics	Energy pyramid builder: set producer biomass; each trophic level is 10% of the one below; pyramid shrinks with arrows showing loss
Human Reproduction	Gametogenesis diagram: spermatogenesis and oogenesis side by side with cell division stages shown in a parallel timeline
Evolution – Natural Selection	Population colour-frequency simulation: add predators; favoured colour variant population rises across generations in a live bar chart

Accountancy (Commerce)

 Concept	 Digital Board Experiment / Visualisation
Journal, Ledger & Trial Balance	T-account board: enter a journal entry; debit and credit sides auto-populate on correct ledger T-accounts; trial balance sums live
Financial Statements	Balance sheet builder: drag asset/liability items into correct sections; totals balance or flash red if out of balance

Cash Flow Statement	Three-activity sorter: classify cash flows into Operating / Investing / Financing columns; net change in cash shown at bottom
Partnership Accounts	Profit-sharing calculator: input ratios; drag profit figure; each partner's share auto-distributes on an animated pie chart

Economics

 Concept	 Digital Board Experiment / Visualisation
Demand & Supply – Market Equilibrium	Shift simulator: drag demand/supply curves left/right; new equilibrium price and quantity recalculate and highlight on graph
Elasticity of Demand	Elasticity explorer: adjust price change %; percentage change in quantity shown; Ed value computed; elastic vs inelastic zone shades
National Income – GDP Concepts	Circular flow diagram: tap each flow (consumption, investment, government, net exports); GDP formula builds on screen
Money & Banking – Credit Creation	Credit multiplier chain: input initial deposit and CRR; chain of bank loans visualises; total money supply shown at bottom
Index Numbers	Price index builder: enter base year and current year prices; index computed; inflation rate derived and shown on chart

History (Humanities)

 Concept	 Digital Board Experiment / Visualisation
Indus Valley Civilisation	Excavation map: reveal layers by dragging a brush; artefacts appear at different depths with date and significance labels
Colonialism & Resistance	Territory map: animate British expansion across India decade by decade; resistance events marked with event-card pop-ups
Partition of India 1947	Radcliffe Line animator: boundary draws over Punjab/Bengal map; demographic data of districts shown alongside border placement
Constitutional Debates	Constituent Assembly: seat map of 299 members; click a cluster to see that group's key argument in the debate speech bubble

Geography (Humanities)

 Concept	 Digital Board Experiment / Visualisation
Earth's Interior & Landforms	Core-to-crust cross-section: zoom in through crust → mantle → outer core → inner core; temperature/density values at each layer
Atmosphere & Climate	Pressure belt diagram: drag latitude slider; Hadley / Ferrel / Polar cells rotate; prevailing winds animate and ITCZ shifts
Hydrological Cycle	Full water cycle canvas: tap Sun to evaporate, clouds form, precipitation falls, infiltration/runoff arrows activate — each process labelled

Population Distribution	India choropleth map: colour intensity by population density; play a year slider 1951–2021; watch density deepen over decades
Human Development Index	HDI component radar chart: input education, income, health scores; HDI value computes and countries ranked on an interactive list

Political Science (Humanities)

 Concept	 Digital Board Experiment / Visualisation
Federalism in India	Powers distribution board: drag policy areas (education, defence, forests) to Union/State/Concurrent lists; errors flag immediately
Election Process in India	Step-by-step election flowchart: Nomination → Manifesto → Campaign → Voting → Counting → Declaration; each stage has video clip link
International Relations	World map with UN bloc overlay: tap country to see vote on a resolution; colour-code Yes/No/Abstain; geopolitical groups cluster

Computer Science (Cl. 11–12)

 Concept	 Digital Board Experiment / Visualisation
Python Programming – Functions	Code editor with step-through debugger: execute line by line; variable values shown in a panel beside code; call stack visualised
Data Structures – Stack, Queue, Linked List	Animated structure: push/pop stack (plates), enqueue/dequeue queue (ticket line), insert/delete linked list nodes in real time
Sorting Algorithms	Bar-chart sorter: watch bubble / insertion / merge / quick sort rearrange coloured bars step-by-step with comparison count shown
Database & SQL	Visual SQL builder: drag table cards, define relationships; write SELECT query; results appear as a highlighted filtered table
Networking – OSI Model	Packet journey: tap 'Send'; data packet shown moving layer by layer (Application → Physical); encapsulation headers added at each step
Artificial Intelligence Basics (Cl. 11+)	Decision tree builder: input training data rows; tree branches draw themselves; classify a new input by walking down branches

RECOMMENDED DIGITAL TOOLS

Tool / Platform	Best Used For
PhET Simulations (phet.colorado.edu)	Physics, Chemistry experiments — class 6 to 12 (free, Hindi available)
GeoGebra (geogebra.org)	All Mathematics concepts — class 6 to 12: graphs, geometry, algebra, calculus
Desmos (desmos.com)	Function graphing, statistics visualisation — class 9 to 12 Maths
OLabs (olabs.edu.in)	NCERT-aligned virtual labs — Biology, Chemistry, Physics class 9–12
DIKSHA (diksha.gov.in)	NCERT curriculum-mapped content, e-books, videos — all classes
Scratch (scratch.mit.edu)	Coding, algorithm concepts — class 6 to 8 Computer Science
Khan Academy (khanacademy.org/intl/hi)	All subjects, Hindi support, adaptive exercises — class 3 to 12
Tinkercad (tinkercad.com)	3D geometry, circuits — class 8 to 12
Google Earth / Maps	Geography, map work — class 6 to 12 Social Science
NCERT ePathshala (epathshala.nic.in)	Official NCERT books, audio-visual — all classes all subjects